Field Dynamics and Environment Modeling

Standalone Report for MSc Thesis:
"Modern Control Methods for Agricultural Irrigation"

Tara Torbati
ITMO University, Saint Petersburg, Russia

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1 Study Site Description

The study site is a 6-hectare agricultural field located in Gilan Province, Iran (38.298°N, 48.847°E), in the humid subtropical coastal zone south of the Caspian Sea. Gilan is Iran's primary rice-producing province, accounting for over 238,000 hectares of paddy fields [1]. The field exhibits a bowl-shaped topography with elevations ranging from 74 m to 181 m above sea level. A digital elevation model (DEM) was obtained from USGS Earth Explorer and processed in QGIS to extract a 10×13 grid of 130 computational agents, each representing approximately 462 m^2.

1.1 Topological Graph Construction

The DEM was used to construct a directed graph encoding water flow topology. For each agent n, the normalized elevation was computed as $\gamma_n = (z_n - z_{\min})/(z_{\max} - z_{\min})$. Each agent sends lateral water flow to all eight-connected neighbors with strictly lower normalized elevation. The number of lower neighbors $N_r(n)$ determines the agent's drainage capacity. Agents with $N_r = 0$ are classified as *sink agents*—local minima where water accumulates. On this bowl-shaped field, sink agents are concentrated near the center and represent areas at risk of waterlogging.

2 Climate Data

2.1 Data Source and Period

Daily meteorological data were obtained from the NASA POWER database for 2000–2025 at coordinates (38.30°N, 48.85°E), providing satellite-derived and reanalysis-based parameters at ∼50×60 km resolution. Thirteen variables were extracted: T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, RH2M, WS2M, PS, T2MDEW, ALLSKY_SFC_SW_DWN, GWETROOT, and GWETTOP.

2.2 Preprocessing

The dataset was filtered to April–October for each year, yielding 5,564 daily records. No missing values were found. One rainfall anomaly was identified (April 7, 2022: 375 mm) and replaced with the calendar-day average (1.2 mm). Two reference evapotranspiration values were computed for each day (Section 2.3).

2.3 Reference Evapotranspiration

Two methods were compared:

Hargreaves equation [2]: $\mathrm{ET}_0^H = 0.0023 \cdot (T_{\mathrm{mean}} + 17.8) \cdot (T_{\max} - T_{\min})^{0.5} \cdot R_s$

FAO Penman-Monteith [2]:

$$\mathrm{ET}_0^{PM} = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{900}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \tag{1}$$

Over the 25-year rice season, Hargreaves overestimated ET_0 by 34% (6.78 vs 5.02 mm/day, $r = 0.965$). This is due to Gilan's high humidity (∼64%) and low wind (∼1.40 m/s), which

suppress evapotranspiration but are not captured by Hargreaves. Penman-Monteith was selected as the standard [2].

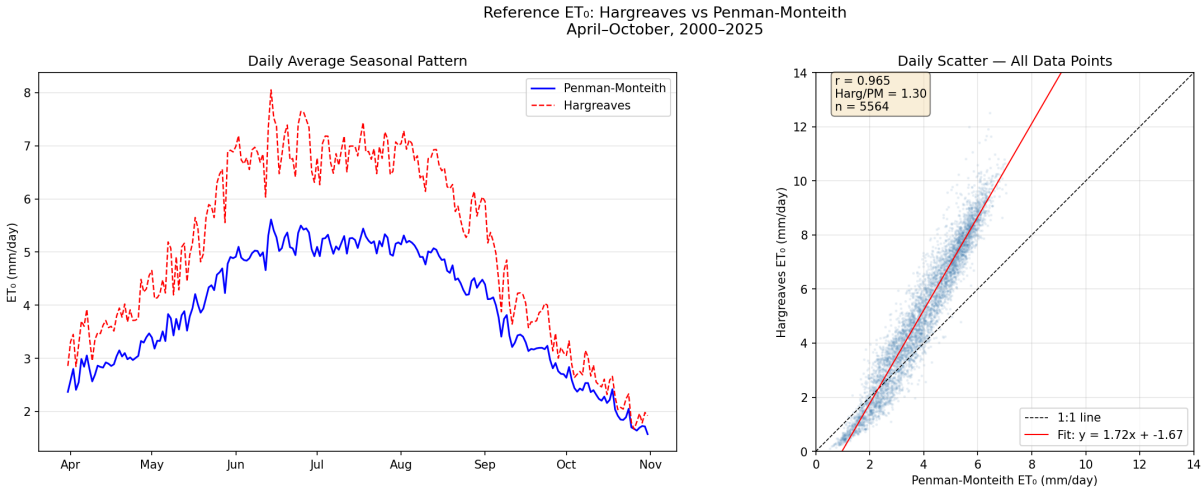

Figure 1: Comparison of Hargreaves and Penman-Monteith ET_0. Left: seasonal pattern. Right: scatter plot with regression.

2.4 25-Year Climatology

Mean temperature ranges from 12°C in early April to 26°C in mid-July. Precipitation shows a dry mid-season (June–August, 0.5–1.5 mm/day) flanked by wetter months. Solar radiation peaks at 26 MJ/m^2/day in June. Relative humidity remains 60–75% throughout—a defining feature of Gilan's Caspian climate. Additional climate figures are provided in Appendix A.

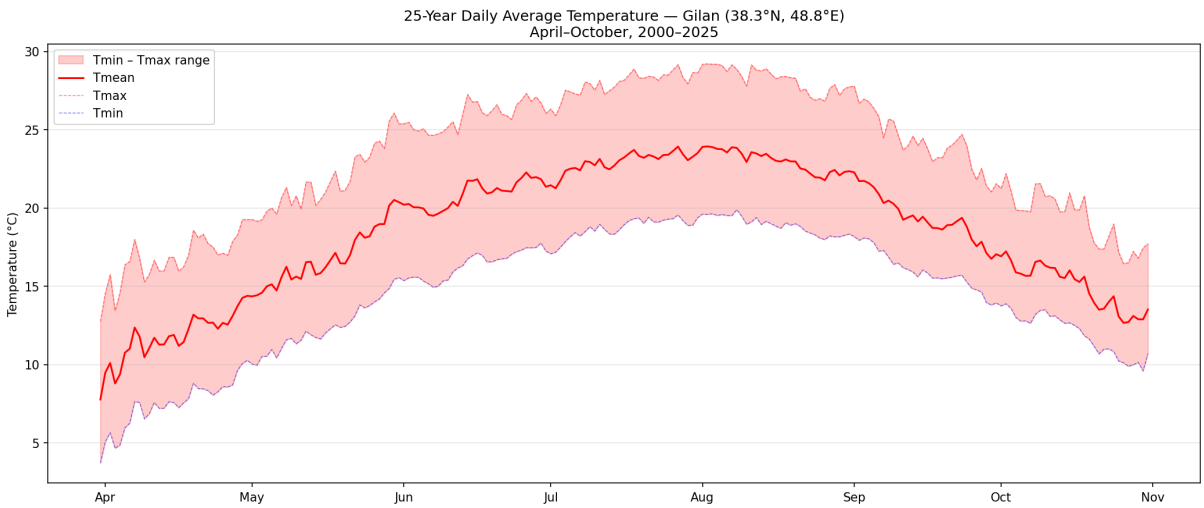

Figure 2: 25-year daily average temperature (April–October, 2000–2025).

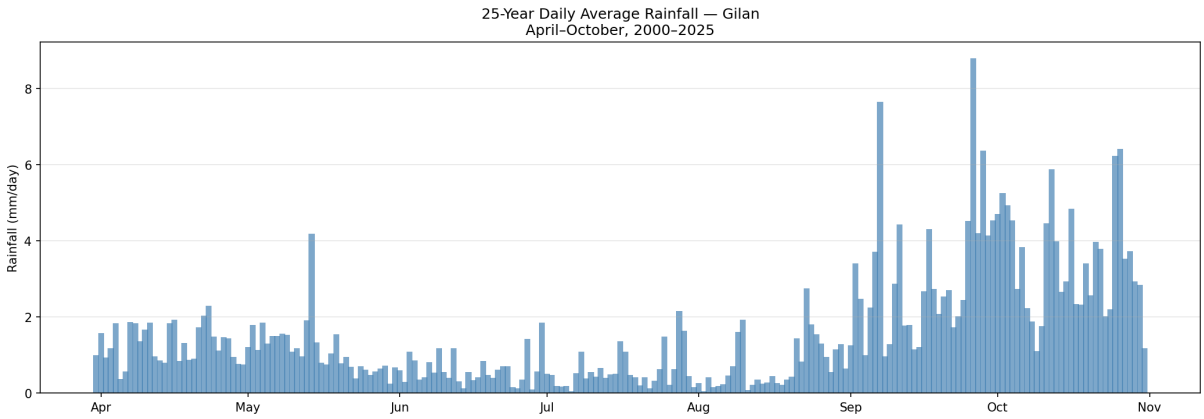

Figure 3: 25-year daily average rainfall (April–October, 2000–2025).

3 Crop Parameterization

Two crops were selected: rice (Hashemi cultivar) and tobacco (*Nicotiana tabacum*). Both are commercially significant in Gilan [1, 3]. Their contrasting characteristics—high K_c/shallow roots/drought-sensitive rice versus moderate K_c/deep roots/drought-tolerant tobacco—enable evaluation of control algorithms under different agronomic conditions.

3.1 Rice (Hashemi Cultivar)

3.1.1 Base Temperature and Thermal Time

The base temperature $T_{\text{base}} = 10°\text{C}$ follows the standard GDD convention [4, 5] and the baseline model [6]. The AquaCrop convention of 8°C [20] produces proportionally higher GDD values without changing maturity dates.

The maturity threshold $\theta_{18} = 1250$ °C·day was calibrated from the 25-year dataset and validated against Sadidi Shal et al. (2021), who measured 1216–1352 GDD for Hashemi rice in Gilan [7]. With a 20-day nursery contributing ∼210 GDD (+5°C greenhouse boost), 20/26 years reach maturity by August 20.

3.1.2 Season Window

Transplanting: May 20 (DOY 141), when 7-day rolling $T_{\text{mean}} \geq 18°\text{C}$ and $T_{\min} \geq 12°\text{C}$ [8, 9].

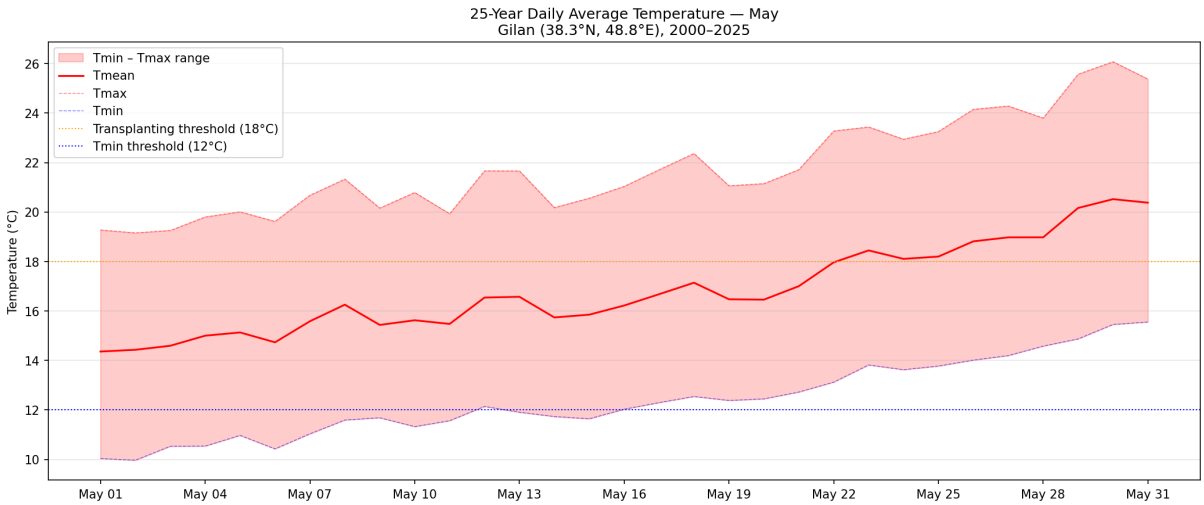

Figure 4: 25-year average May temperature with transplanting thresholds.

Harvest: August 20 (DOY 233). WMO R20mm events (≥20 mm/day) [10] increase sharply after this date, causing lodging [10] and harvest disruption [11].

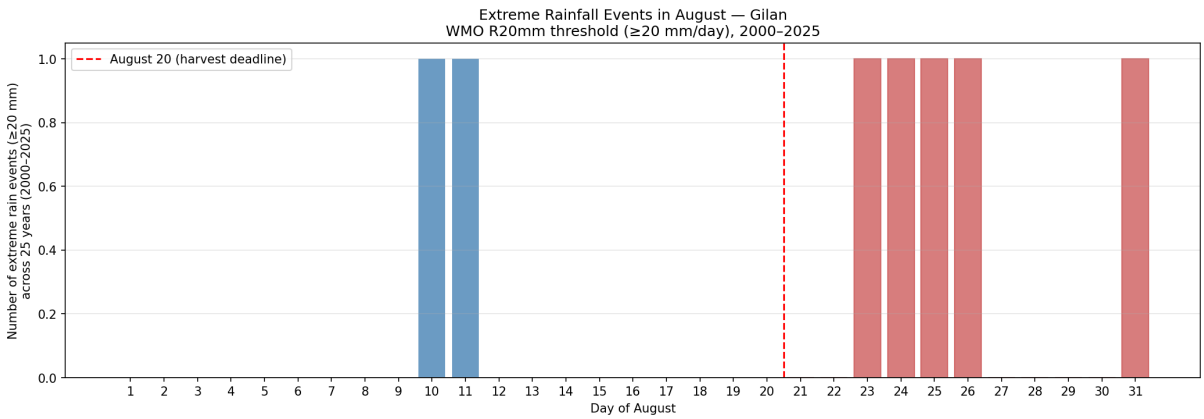

Figure 5: Extreme rainfall events (≥20 mm/day) per day of August, 2000–2025.

Field season: 93 days. Nursery provides $x_{2,\text{init}} = 210$ GDD. Other parameters: $K_c = 1.15$, $p = 0.20$ [2], $\theta_5 = 400$ mm, $\theta_{13} = 0.65$ g/MJ [12, 13], HI = 0.42 [14]. Expected yield: 3500–4000 kg/ha (optimized) or ∼2000 kg/ha (traditional flooding, consistent with local farmer reports).

3.2 Tobacco (*Nicotiana tabacum*)

3.2.1 Base Temperature and Thermal Time

$T_{\text{base}} = 10°$C, within the 10–13°C restriction range [15]. $\theta_{18} = 1200$ GDD (field only), calibrated to match the 104-day season. Duration consistent with FAO (90–120 days) [16] and Britannica (100–130 days) [17]. $x_{2,\text{init}} = 0$ (nursery conditions unknown—seedlings delivered by company).

3.2.2 Season Window

Transplanting: May 25 (DOY 146). Harvest: September 5 (DOY 249). 104 field days.

3.2.3 Other Parameters

θ_5 = 700 mm (FAO: 500–1000 mm) [16]. $K_c = 0.90$, $p = 0.50$ [16]. θ_9 = 30°C (FAO optimal ceiling) [15, 16]. θ_{13} = 0.30 g/MJ [18], HI = 0.55. Expected yield: 2000–2500 kg/ha [16].

3.3 Parameter Summary

Table 1: Complete crop and soil parameters.

Parameter	Symbol	Rice	Tobacco	Source
Shared soil parameters (silty loam)				
Water uptake coeff.	θ_1	0.096	0.096	[6]
Wilting point (vol.)	θ_2	0.15	0.15	[2]
SCS threshold	θ_3	10 mm	10 mm	[6]
Drainage coeff.	θ_4	0.05	0.05	[6]
Field capacity (vol.)	θ_6	0.35	0.35	[2]
Crop-specific parameters				
Root depth	θ_5	400 mm	700 mm	[2], [16]
Base temperature	θ_7	10°C	10°C	[5], [15]
Heat stress onset	θ_9	35°C	30°C	[16]
Extreme heat	θ_{10}	42°C	38°C	–
RUE	θ_{13}	0.65	0.30	[12], [18]
Drought sensitivity	θ_{14}	0.80	0.60	[6]
Maturity GDD	θ_{18}	1250	1200	[7], calib.
Max interception	θ_{19}	0.95	0.90	[6]
50% intercept. GDD	θ_{20}	417	450	[19]
Crop coefficient	K_c	1.15	0.90	[2], [16]
Depletion fraction	p	0.20	0.50	[2], [16]
Harvest index	HI	0.42	0.55	[14], [16]
Nursery GDD	$x_{2,\text{init}}$	210	0	Calib.
Initial biomass	$x_{4,\text{init}}$	60 g/m^2	8 g/m^2	Est.
Season	–	DOY 141–233	DOY 146–249	This study
Field days	–	93	104	–
FC	–	140 mm	245 mm	$\theta_6 \times \theta_5$
WP	–	60 mm	105 mm	$\theta_2 \times \theta_5$
TAW	–	80 mm	140 mm	FC − WP
RAW	–	16 mm	70 mm	$p\times$ TAW

4 Water Budget and Scenario Selection

4.1 Water Budget Formulation

Crop water demand: $\text{ET}_c = \overline{\text{ET}_0^{PM}} \cdot K_c \cdot N_{\text{days}}$. Full irrigation need: $I_{\text{full}} = \max(\text{ET}_c - \overline{P}, 0)$, using 25-year average rainfall (Method B).

Table 2: Water budget summary.

	Rice	**Tobacco**
Season duration	93 days	104 days
Avg ET_0 (PM)	5.02 mm/day	4.95 mm/day
Crop demand	537.0 mm	463.4 mm
Avg season rainfall	53.3 mm	74.5 mm
Full irrigation need	483.7 mm	389.0 mm
Budget at 70%	338.6 mm	272.3 mm
Budget at 50%	241.8 mm	194.5 mm

Rice demand (537 mm) is consistent with Hashemi field measurements: 534 mm (SRI) to 632 mm (traditional flooding) [14]. Tobacco demand (463 mm) falls within the FAO range of 400–600 mm [16]. Despite a longer season, tobacco needs less irrigation due to lower K_c and higher rainfall.

4.2 Scenario Selection

Table 3: Selected climate scenarios.

Scenario	**Year**	**Rice (mm)**	**Tobacco (mm)**
Dry	2022	39.7	54.0
Moderate	2020	42.1	73.3
Wet	2024	176.8	163.2

5 Agent-Based Model

The ABM follows López-Jiménez et al. [6]. Each of $N = 130$ agents maintains: x_1 (soil water, mm), x_2 (thermal time, °C·day), x_3 (maturity stress), x_4 (biomass, g/m^2).

5.1 State Equations

Water balance:

$$x_1^n(k+1) = x_1^n(k) - \varphi_1^n - \varphi_2^n - \varphi_3^n - \varphi_{4,\text{out}}^n + \varphi_{4,\text{in}}^n + \varphi_{2,\text{in}}^n + P(k) + u^n(k) \tag{2}$$

Thermal time: $x_2^n(k+1) = x_2^n(k) + \max(T_{\text{mean}} - \theta_7, 0)$
Maturity stress: $x_3^n(k+1) = x_3^n(k) + \theta_{11}(1 - h_2) + \theta_{12}(1 - h_3^n)$
Biomass:

$$x_4^n(k+1) = x_4^n(k) + \theta_{13} \cdot h_3^n \cdot h_6^n \cdot h_7 \cdot g(x_2^n) \cdot R_s(k) \tag{3}$$

5.2 Water Fluxes

$\varphi_1^n = \min(\max(\theta_1(x_1^n - \theta_2\theta_5), 0), \text{ET}_0)$ (transpiration)
$\varphi_2 = (P - \theta_3)^2/(P + 4\theta_3)$ if $P > \theta_3$, else 0 (SCS runoff)
$\varphi_3^n = \theta_4 \cdot \max(x_1^n + P - \varphi_2 - \theta_6\theta_5, 0)$ (deep drainage)
$\varphi_{4,\text{out}}^n = \max(x_1^n - \theta_6\theta_5, 0)$; $\varphi_{4,\text{in}}^m = \sum_{n \in \mathcal{U}(m)} \varphi_{4,\text{out}}^n / N_r(n)$ (lateral flow)

5.3 Stress and Growth Functions

$h_1 = \max(T_{\text{mean}} - \theta_7, 0)$; h_2: linear between θ_9 (no stress) and θ_{10} (full stress); $h_3^n = 1-\theta_{14}\cdot\max(1-\varphi_1^n/\text{ET}_0, 0)$ (drought); $h_6^n = \text{clip}(1-(x_1^n-\theta_6\theta_5)/(\theta_6\theta_5), 0, 1)$ (waterlogging); $h_7 = 1$ if $T_{\text{mean}} > \theta_7$, else 0.

Growth function:

$$g(x_2^n) = \begin{cases} \theta_{19}/(1+e^{-0.01(x_2^n-\theta_{20})}) & x_2^n \leq \theta_{18}/2 \\ \theta_{19}/(1+e^{0.01(x_2^n+x_3^n-\theta_{18})}) & x_2^n > \theta_{18}/2 \end{cases} \tag{4}$$

6 Model Corrections and Extensions

Three corrections to the baseline [6] were implemented.

6.1 Mass Conservation

The original water balance includes $\varphi_{4,\text{in}}$ but omits $\varphi_{4,\text{out}}$, violating mass conservation. The corrected Eq. 2 includes $-\varphi_{4,\text{out}}^n$.

6.2 Drought Stress on Biomass

The original biomass equation lacks h_3, so drought does not reduce biomass. The corrected Eq. 3 includes h_3^n, reducing biomass by up to $\theta_{14} \times 100\%$ under maximum drought (80% for rice, 60% for tobacco). This is essential for irrigation optimization: without it, the controller has no incentive to irrigate.

6.3 Surface Runoff Redistribution

Originally, φ_2 leaves the system. A simple redistribution was implemented: each agent's runoff flows to lower neighbors with immediate infiltration ($\varphi_{2,\text{in}}^n$). A cascading alternative (top-to-bottom SCS re-application) was also tested.

Table 4: Runoff redistribution comparison, no irrigation, wet scenario (2024).

Mode	**Yield (kg/ha)**	x_1 **sinks (mm)**	x_1 **hills (mm)**
None (original)	2135	71.2	71.2
Simple (A)	2273	74.8	72.0
Cascade (B)	2278	79.5	72.0

Simple redistribution captures the first-order effect (6.5% yield improvement) with negligible difference from cascade (0.2%). Simple was adopted.

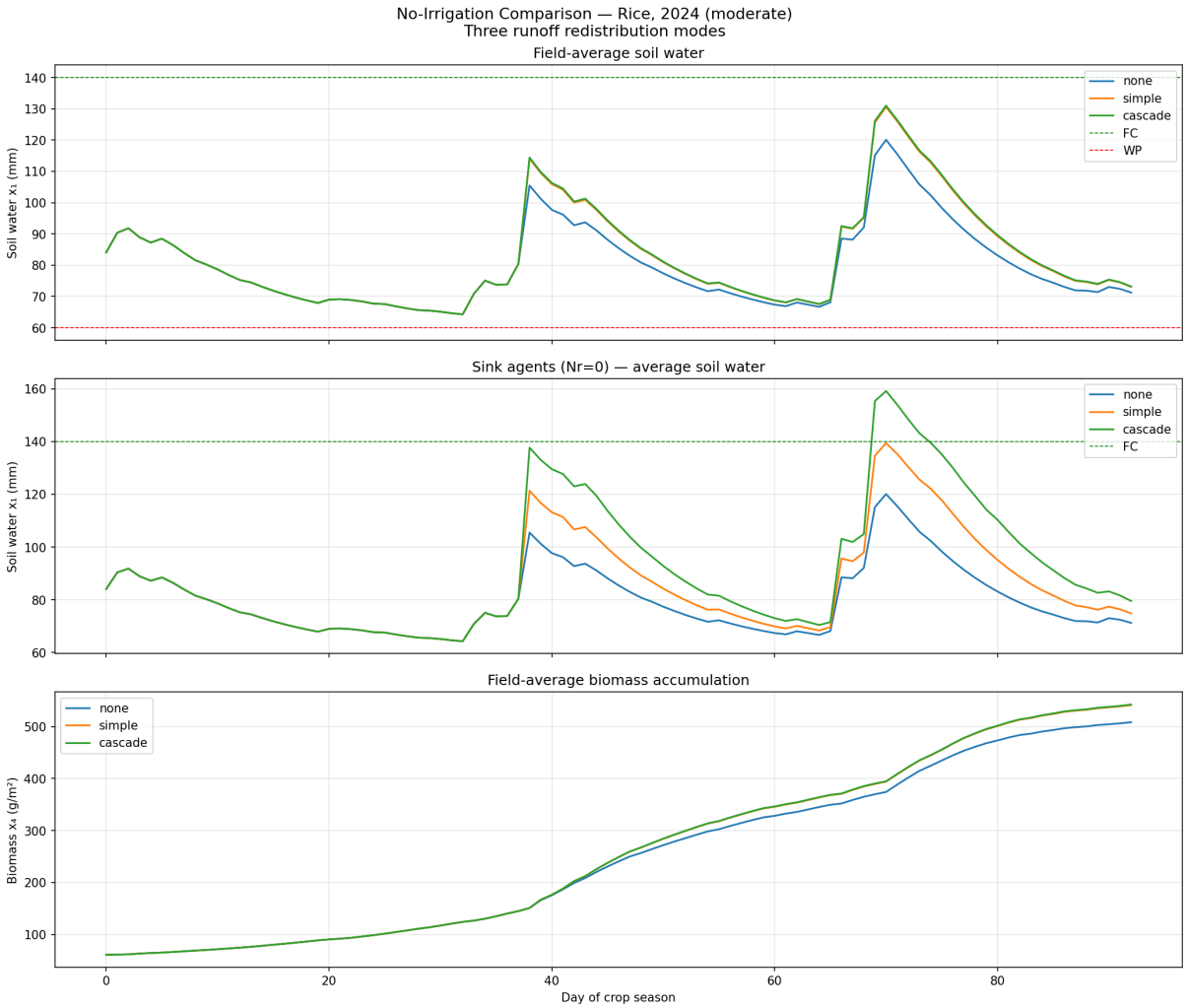

Figure 6: No-irrigation runoff mode comparison, wet scenario (2024).

6.4 Biomass Calibration

θ_{13} was calibrated so x_4 represents g/m^2 dry matter. Yield: $Y = x_4 \times \mathrm{HI} \times 10$ (kg/ha). Rice: validated against ∼2000 kg/ha (farmer reports) and 3500–4000 kg/ha (optimized) [14]. Initial conditions: rice $x_{2,\mathrm{init}} = 210$, $x_{4,\mathrm{init}} = 60$; tobacco $x_{2,\mathrm{init}} = 0$, $x_{4,\mathrm{init}} = 8$.

7 Model Validation

7.1 GWETROOT Cross-Validation

ABM soil water was compared to NASA MERRA-2 GWETROOT. Both normalized: ABM as $(x_1 - \mathrm{WP})/(\mathrm{FC} - \mathrm{WP})$; GWETROOT as native 0–1 fraction. Pearson correlation assessed temporal pattern agreement.

Table 5: Cross-validation: ABM vs NASA GWETROOT.

Scenario	**Year**	**Pearson** r	p**-value**
Dry	2022	0.738	3.1×10^{-17}
Moderate	2020	0.534	3.4×10^{-8}
Wet	2024	0.337	9.6×10^{-4}

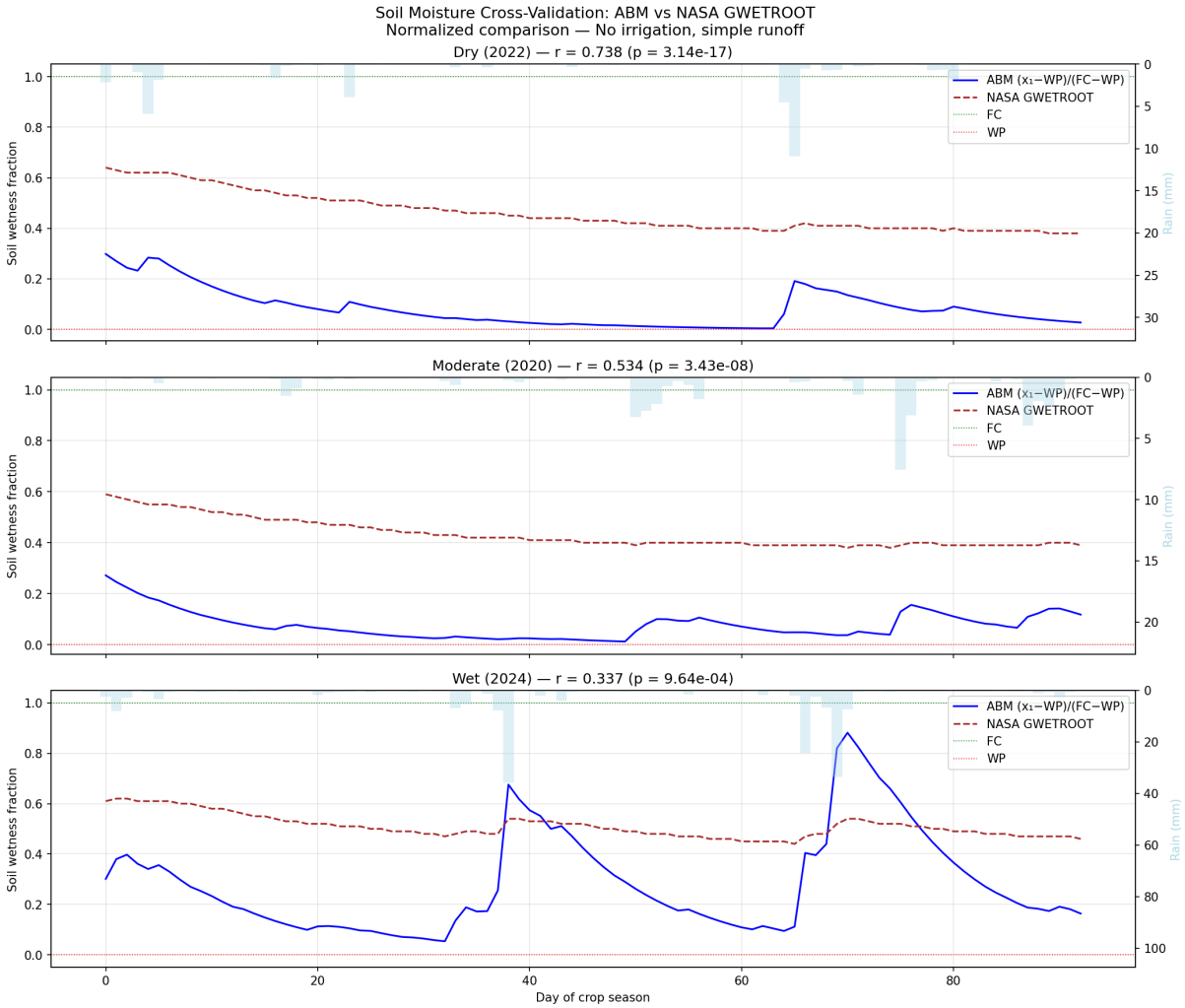

Figure 7: Normalized soil moisture: ABM (blue) vs NASA GWETROOT (brown).

The dry-year $r = 0.74$ ($p < 0.001$) confirms the ABM captures wetting/drying dynamics. Decreasing correlation with increasing rainfall is expected (reduced signal contrast in wet conditions). The absolute level difference (ABM ∼0.08 vs GWETROOT ∼0.46) reflects: (1) NASA's deeper root zone (∼1000 mm vs 400 mm); (2) GWETROOT observes irrigated fields while the ABM simulates rain-fed conditions; (3) different soil parameterizations.

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A Supplementary Climate Figures

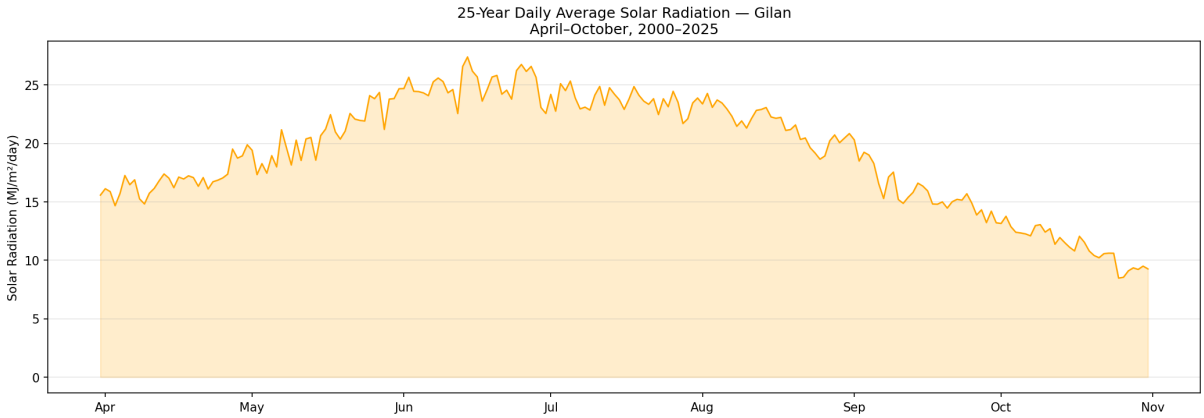

Figure 8: 25-year daily average solar radiation, April–October.

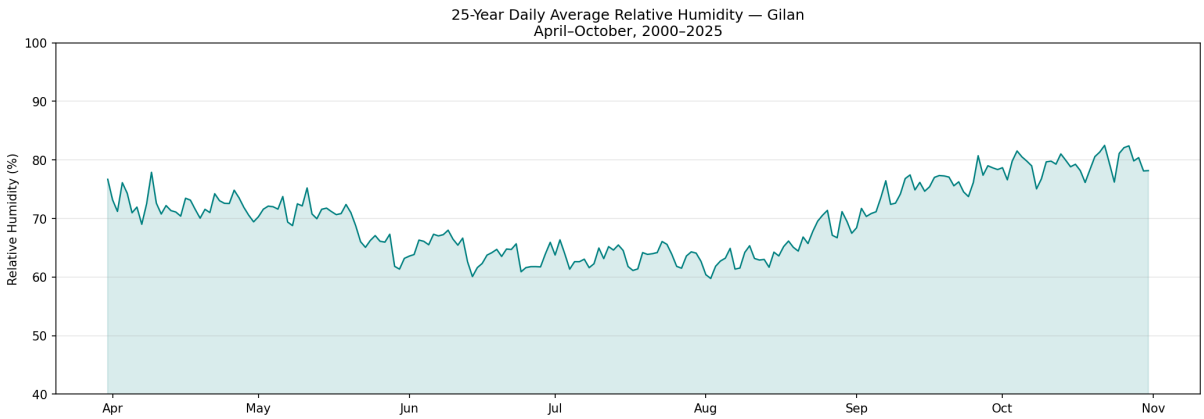

Figure 9: 25-year daily average relative humidity, April–October.

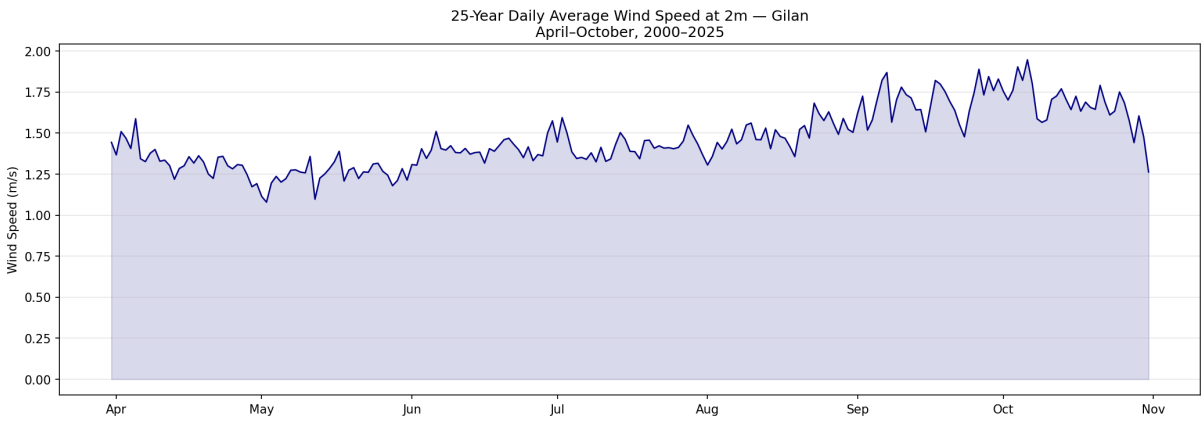

Figure 10: 25-year daily average wind speed, April–October.

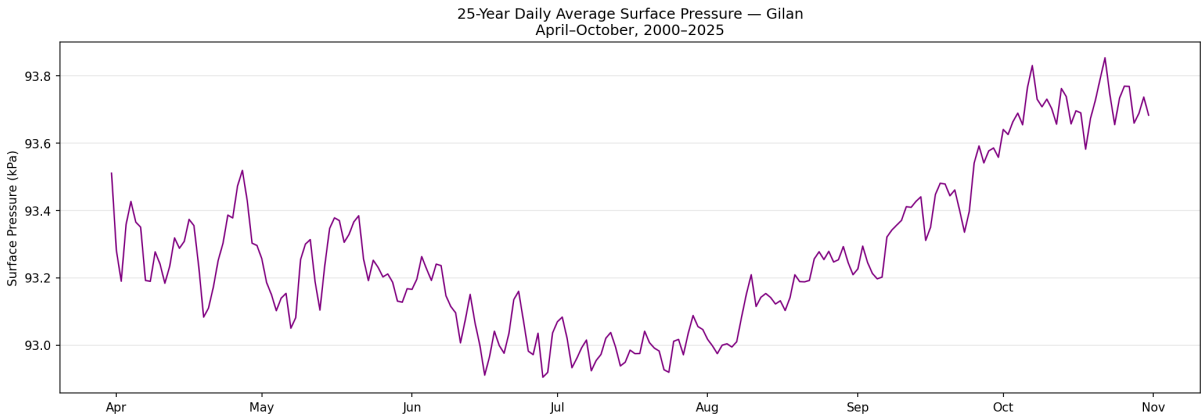

Figure 11: 25-year daily average surface pressure, April–October.

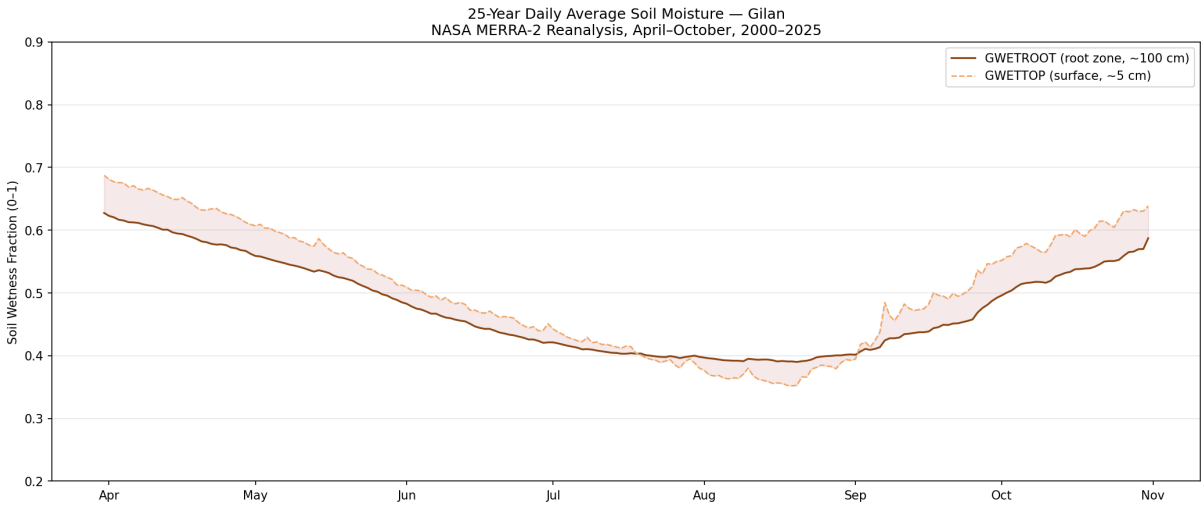

Figure 12: 25-year daily average NASA MERRA-2 soil wetness, April–October.

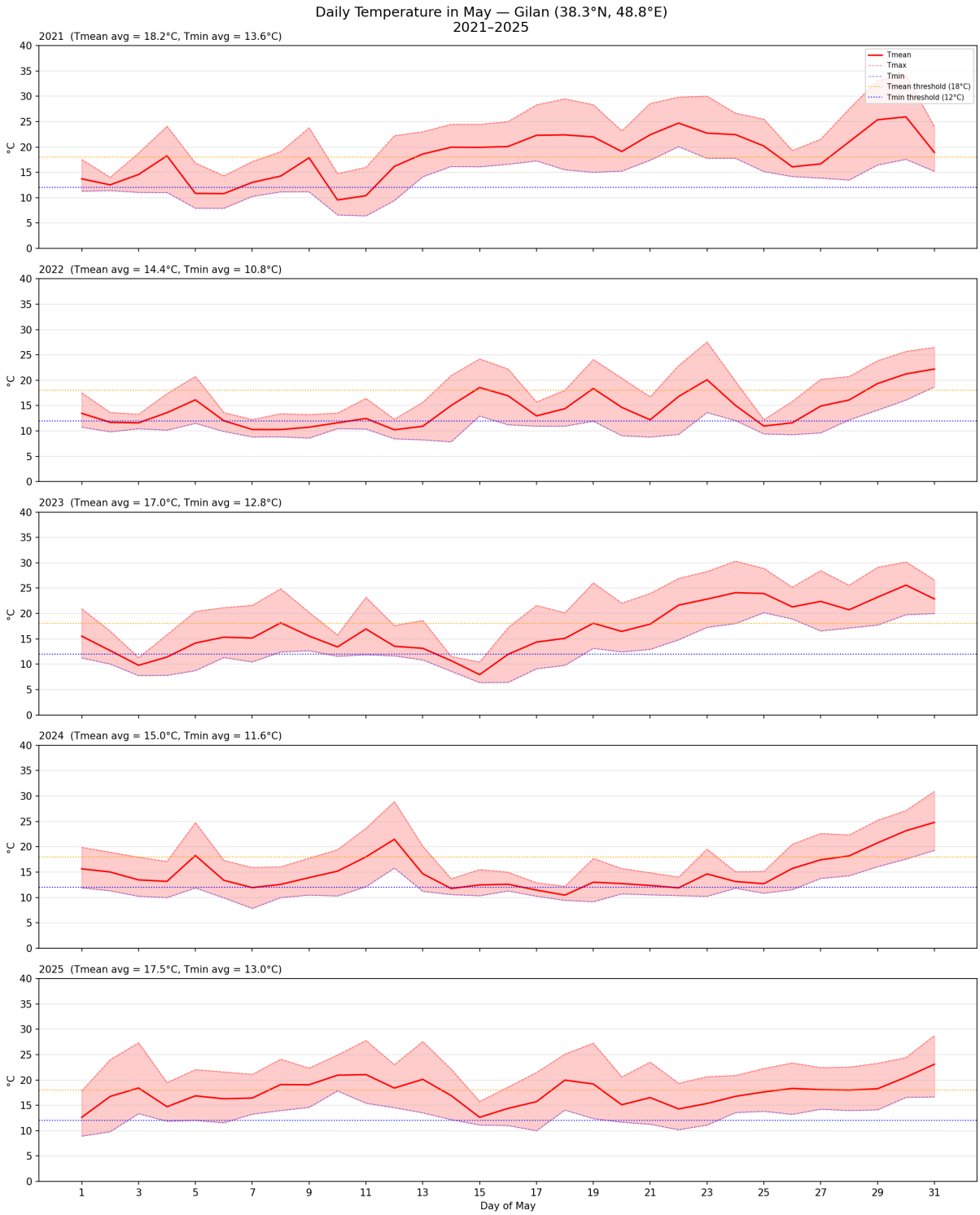

Figure 13: Daily temperature in May for 2021–2025.

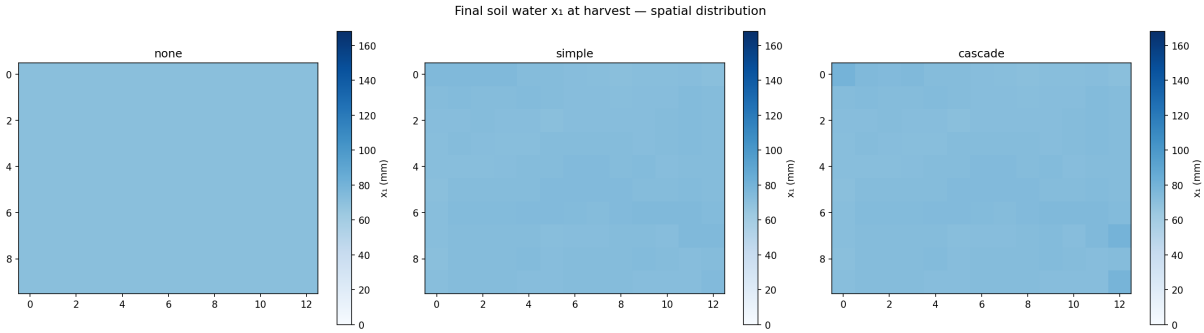

Figure 14: Spatial distribution of x_1 at harvest, three runoff modes, wet scenario (2024).

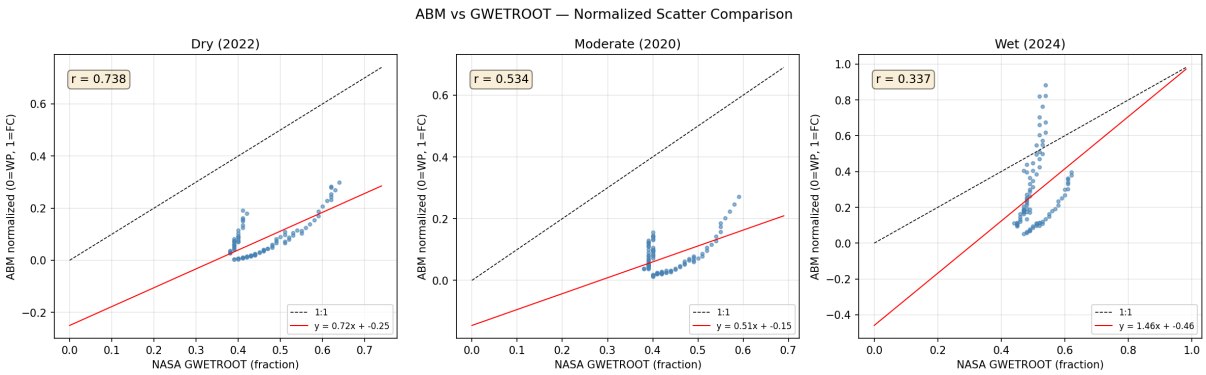

Figure 15: Scatter plots: ABM vs NASA GWETROOT for three scenarios.